Part I Exam Cheatsheets: Probability and Statistics Theory

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1 Common distributions

Distribution	Mean	Variance	PMF/PDF
Unif $([a,b])$	$\frac{a+b}{2}$	$\frac{(b-a)(b-a+2)}{12}$	$\frac{1}{b-a+1}$
Ber (p)	p	$p(1-p)$	$p^x(1-p)^{1-x}$
Bin (n,p)	np	$np(1-p)$	$\binom{n}{k}p^k(1-p)^{n-k}$
Geom (p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$p(1-p)^{k-1}$
Poiss (λ)	λ	λ	$\frac{\lambda^k e^{-\lambda}}{k!}$
Unif $([a,b])$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{1}{b-a}$
$\mathcal{N}(\mu,\sigma^2)$	μ	σ^2	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
Exp (λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\lambda e^{-\lambda x}$
Beta (α,β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$
Gamma (α,β)	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$

2 MLE and Log-Likelihood

Dist	$\log p(x\|\theta)$	$\ell'(\theta)$	$\ell''(\theta)$	MLE
Bern(p)	$x\log p+(1-x)\log(1-p)$	$\frac{x}{p}-\frac{1-x}{1-p}$	$-\frac{x}{p^2}-\frac{1-x}{(1-p)^2}$	$\hat{p}=\bar{x}$
Bin(n,p)	$k\log p+(n-k)\log(1-p)$	$\frac{k}{p}-\frac{n-k}{1-p}$	$-\frac{k}{p^2}-\frac{n-k}{(1-p)^2}$	$\hat{p}=\frac{k}{n}$
Geom(p)	$\log p+(k-1)\log(1-p)$	$\frac{1}{p}-\frac{k-1}{1-p}$	$-\frac{1}{p^2}-\frac{k-1}{(1-p)^2}$	$\hat{p}=\frac{1}{\bar{x}}$
Pois(λ)	$x\log\lambda-\lambda$	$\frac{x}{\lambda}-1$	$-\frac{x}{\lambda^2}$	$\hat{\lambda}=\bar{x}$
Exp(λ)	$\log\lambda-\lambda x$	$\frac{1}{\lambda}-x$	$-\frac{1}{\lambda^2}$	$\hat{\lambda}=\frac{1}{\bar{x}}$
$\mathcal{N}(\mu,\sigma^2)$	$-\frac{(x-\mu)^2}{2\sigma^2}-\frac{1}{2}\log\sigma^2$	$\frac{x-\mu}{\sigma^2}$	$-\frac{1}{\sigma^2}$	$\hat{\mu}=\bar{x}$

3 Conditioning, independence, and counting

3.1 Conditional probability

$$\mathbb{P}\left(\bigcap_{i=1}^{n}A_i\right)=\mathbb{P}(A_1)\prod_{i=2}^{n}\mathbb{P}\left(A_i\mid\bigcap_{j=1}^{i-1}A_j\right)$$

$$\mathbb{P}(B)=\sum_i\mathbb{P}(A_i)\,\mathbb{P}(B\mid A_i),\quad \mathbb{P}(A_i\mid B)=\frac{\mathbb{P}(B\mid A_i)\,\mathbb{P}(A_i)}{\sum_j\mathbb{P}(B\mid A_j)\,\mathbb{P}(A_j)}$$

$$\mathbb{P}(A\mid B)=\frac{\mathbb{P}(B\mid A)\,\mathbb{P}(A)}{\mathbb{P}(B\mid A)\,\mathbb{P}(A)+\mathbb{P}(B\mid A^c)\,\mathbb{P}(A^c)}$$

3.2 Independence

$$\mathbb{P}\left(\bigcap_{i\in S}A_i\right)=\prod_{i\in S}\mathbb{P}(A_i)$$

- $A\perp B\iff\mathbb{P}(A\mid B)=\mathbb{P}(A)$
- Independence $\not\Rightarrow$ conditional independence
- $X\perp Y\Rightarrow\mathbb{E}[g(X)h(Y)]=\mathbb{E}[g(X)]\,\mathbb{E}[h(Y)]$

3.3 Counting

- Permutations: $n!$
- k-permutations: $n!/(n-k)!$
- Combinations: $n!/(k!(n-k)!)$
- Partitions: $n!/(n_1!\cdots n_r!)$

4 Random variables

4.1 Expectation and variance

$$\mathbb{E}[aX+bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y], \quad \mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] = \int \mathbb{E}[X \mid Y=y] f_Y(y)\,dy$$

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2, \quad \operatorname{Var}(cX) = c^2 \operatorname{Var}(X)$$

$$\operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y], \quad \operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y)$$

$$\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y]), \quad X \perp Y \Rightarrow \operatorname{Cov}(X,Y) = 0$$

4.2 Conditioning rules

$$\mathbb{E}[g(Y)X \mid Y] = g(Y)\,\mathbb{E}[X \mid Y], \quad \operatorname{Var}(X \mid Y) = \mathbb{E}[X^2 \mid Y] - (\mathbb{E}[X \mid Y])^2$$

$$X \perp Y \Rightarrow \mathbb{E}[X \mid Y] = \mathbb{E}[X], \quad \operatorname{Var}(X \mid Y) = \operatorname{Var}(X)$$